Pandrosion Operator and Smale's 17th Problem

Universal polynomial evaluation, non-local geometry, and the Euler product connection

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Abstract

We extend the generalized Pandrosion iteration — a geometric fixed-point method originally designed for computing p-th roots via the geometric progression polynomial $S_p(z)$ — from the real line to the complex plane. We show that the Pandrosion sequence is a special case of a *universal evaluation generator* for any complex polynomial $P(z)$: by anchoring the difference quotient $Q(z_0, z) = (P(z) - P(z_0))/(z - z_0)$, we define a non-local rational iteration $z_{n+1} = z_0 - P(z_0)/Q(z_0, z_n)$ that recovers the exact p-th root formula and admits Newton's method as the limiting case where the anchor tracks the iterate. For $P(z) = xz^p - 1$ we characterize the complex contraction ratio and show that its divergence region strictly isolates near the negative real axis.

Our main rigorous contributions concern the *adaptive-anchor* Pandrosion–Steffensen scheme and are threefold: (i) a universal local convergence theorem establishing an explicit contraction basin of radius $\eta_P(\zeta)$ around every simple root; (ii) a pole-avoidance theorem showing that the set of initial data whose orbit ever lands on a pole of Q has four-dimensional Lebesgue measure zero; (iii) a post-lock-in complexity bound of $O(\log\log\varepsilon^{-1})$ Steffensen steps. We formulate a global full-measure convergence conjecture consistent with the observed 100% grid-convergence but beyond our proof techniques, and we discuss precisely why this leaves Smale's 17th problem open. We close with a formal Pandrosion-language rewriting of the Euler product for $\zeta(s)$, recording a structural observation while explicitly disclaiming any implication for the Riemann Hypothesis.

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1 Introduction

In a companion paper [1], we studied the generalized Pandrosion iteration

$$s_{n+1} = 1 - \frac{x-1}{x \cdot S_p(s_n)}, \qquad S_p(s) = \sum_{k=0}^{p-1} s^k, \qquad v_n = x \cdot s_n^{p-1}, \tag{1}$$

for real $x > 0$ and integer $p \geq 2$. That construction, rooted in Pandrosion of Alexandria's geometric method for duplicating the cube [3], converges linearly to $s^* = x^{-1/p}$ with an explicit contraction ratio $\lambda_{p,x}$ depending only on $\alpha = x^{1/p}$. Steffensen's acceleration transforms the linear method into a quadratically convergent, derivative-free algorithm.

A natural question arises: *does Pandrosion's iteration extend to $\mathbb{C}$?* Nothing in the formula (1) requires x or s to be real — the geometric sum S_p and the rational map $h(s) = 1 - (x-1)/(x \cdot S_p(s))$ are well-defined for complex arguments.

This extension is not merely formal. By moving to $\mathbb{C}$, we uncover foundational properties:

- **General Polynomial Resolution**: Pandrosion's root-finding sequence uniquely categorizes as an anchor-based difference quotient evaluation applicable uniformly across arbitrary polynomials (Smale's 17th problem algorithm).
- **Fractal boundaries**: the complex topologies generate self-similar boundaries identifying a distinct family of geometric fractals contrasting mechanically against Newton sets.

- **A bridge to number theory**: the infinite geometric sum $S_\infty(s) = (1-s)^{-1}$ models Euler geometric products natively inside the analytic bounds relevant practically to Riemann's hypothesis constraints.

Outline. Section 2 defines the elementary complex Pandrosion geometry. Section 3 extends contraction mapping globally. Section 4 presents the critical theoretical advance bridging Pandrosion evaluation toward Smale's 17th universal convergence challenge for arbitrary expressions. Section 5 introduces the adaptive-anchor algorithm and reports the key empirical observation that it achieves 100% grid-convergence on test polynomials. Section 6 develops the rigorous convergence theory: a universal local contraction theorem, a full-measure pole-avoidance theorem, a $O(\log\log\varepsilon^{-1})$ complexity bound, and an honest statement of what remains conjectural. Section 7 identifies non-local dynamics topology. Section 8 establishes analytical quadratic convergence parameters globally. Section 9 establishes structural Euler product identities natively. Section 10 computes localized divergence spaces mathematically. Section 11 resolves algorithmic objectives.

2 Complex Pandrosion iteration

2.1 Definition

Definition 2.1 (Complex Pandrosion iteration)**.** Let $x \in \mathbb{C} \setminus \{0,1\}$ and $p \geq 2$ be an integer. The **complex Pandrosion iteration** is the map $h : \mathbb{C} \to \mathbb{C}$ defined by

$$h(s) = 1 - \frac{x-1}{x \cdot S_p(s)}, \qquad S_p(s) = \sum_{k=0}^{p-1} s^k = \frac{1-s^p}{1-s}. \tag{2}$$

The output sequence is $v_n = x \cdot s_n^{p-1}$.

Remark 2.1. The map h is a rational function of s of degree $p-1$ (after clearing the denominator $S_p(s)$). It has poles at the $(p-1)$-th roots of unity other than $s = 1$ (where S_p vanishes) and a removable singularity at $s = 1$ (where $S_p(1) = p$).

2.2 Fixed points

Theorem 2.1 (Complex fixed points)**.** The fixed points of h in $\mathbb{C}$ are exactly

$$s_k^* = \alpha_k^{-1} = x^{-1/p}\, e^{-2\pi i k/p}, \qquad k = 0, 1, \ldots, p-1, \tag{3}$$

where $\alpha_k = x^{1/p} e^{2\pi i k/p}$ are the p complex p-th roots of x. The corresponding outputs are $v_k^* = x \cdot (s_k^*)^{p-1} = \alpha_k$.

Proof. At a fixed point s^*, we have $h(s^*) = s^*$, i.e.

$$1 - \frac{x-1}{x \cdot S_p(s^*)} = s^* \quad \Longleftrightarrow \quad x \cdot S_p(s^*) \cdot (1 - s^*) = x - 1.$$

Since $S_p(s)(1-s) = 1 - s^p$, this gives $x(1 - (s^*)^p) = x - 1$, hence $(s^*)^p = 1/x$, i.e. s^* is a p-th root of x^{-1}. There are exactly p such roots in $\mathbb{C}$, given by (3). The output formula follows from $v^* = x \cdot (s^*)^{p-1} = x \cdot (x^{-1/p})^{p-1} \cdot e^{-2\pi i k(p-1)/p} = x^{1/p} e^{2\pi i k/p} = \alpha_k$. □

Remark 2.2 (Geometry of fixed points)**.** The p fixed points s_k^* lie on a circle of radius $|x|^{-1/p}$ centered at the origin, equally spaced by angle $2\pi/p$. For real $x > 1$, the principal fixed point $s_0^* = x^{-1/p} \in (0,1)$ is the one studied in [1].

Example 2.1. For $p = 3$, $x = 2 + i$:

$$\begin{aligned}
\alpha_0 &= (2+i)^{1/3} \approx 1.2921 + 0.2013i, \\
\alpha_1 &= \alpha_0\, e^{2\pi i/3} \approx -0.8204 + 1.0183i, \\
\alpha_2 &= \alpha_0\, e^{4\pi i/3} \approx -0.4717 - 1.2196i.
\end{aligned}$$

The three fixed points $s_k^* = \alpha_k^{-1}$ form a triangle in $\mathbb{C}$.

3 Complex contraction ratio

3.1 Extension to $\mathbb{C}$

Theorem 3.1 (Complex contraction ratio)**.** At the principal fixed point $s_0^* = \alpha^{-1}$ with $\alpha = x^{1/p}$ (principal branch), the contraction ratio is

$$\lambda_{p,x} = h'(s_0^*) = \frac{(\alpha - 1)\displaystyle\sum_{k=1}^{p-1} k\,\alpha^{p-1-k}}{\displaystyle\sum_{k=0}^{p-1} \alpha^k}, \qquad \alpha = x^{1/p} \in \mathbb{C}. \tag{4}$$

This is the same formula as in the real case [1], now with $\alpha \in \mathbb{C}$. The iteration converges locally to s_0^* whenever $|\lambda_{p,x}| < 1$.

Proof. The derivative $h'(s) = \frac{x-1}{x} \cdot \frac{S_p'(s)}{S_p(s)^2}$ is a rational function of s. Evaluating at $s_0^* = \alpha^{-1}$ by the same algebraic manipulations as in [1] (which are valid over $\mathbb{C}$) yields (4). □

Remark 3.1. Since $\lambda_{p,x} \in \mathbb{C}$ in general, the convergence may be *spiraling* (when $\mathrm{Im}(\lambda) \neq 0$): the iterates wind around the fixed point while contracting. The convergence rate is governed by $|\lambda_{p,x}|$, not $\lambda_{p,x}$ itself.

3.2 Convergence region

Definition 3.1 (Convergence region)**.** The **convergence region** of the complex Pandrosion iteration is

$$\mathcal{C}_p = \{x \in \mathbb{C} \setminus \{0\} : |\lambda_{p,x}| < 1\}.$$

$\mathrm{Re}(x)$	$\mathrm{Im}(x)$	$\lvert\lambda_{3,x}\rvert$	$\in \mathcal{C}_3$?
0.5	0	0.238	✓
1	1	0.283	✓
2	0	0.220	✓
2	1	0.294	✓
3	2	0.427	✓
5	0	0.468	✓
−1	1	0.874	✓
−1	0	1.323	×
−2	0	1.260	×
−2	2	0.877	✓

Table 1: Modulus of the contraction ratio $|\lambda_{3,x}|$ for various $x \in \mathbb{C}$. Divergence ($|\lambda| \geq 1$) occurs only near the negative real axis.

Proposition 3.1 (Structure of the divergence set)**.** For each $p \geq 2$, the divergence set $\mathcal{D}_p = \mathbb{C} \backslash \mathcal{C}_p$ is a neighborhood of the negative real ray $(-\infty, 0)$. More precisely:

1. $\mathcal{D}_p$ is symmetric about the real axis: $x \in \mathcal{D}_p \Leftrightarrow \bar{x} \in \mathcal{D}_p$.
2. $\mathcal{D}_p$ shrinks as $\mathrm{Im}(x)$ increases: for fixed $\mathrm{Re}(x) < 0$, there exists $b_0 > 0$ such that $|\lambda_{p,x}| < 1$ for $|\mathrm{Im}(x)| > b_0$.
3. The positive real axis $x > 0$ lies entirely in $\mathcal{C}_p$.

Proof. Part (1) follows from $\overline{\lambda_{p,x}} = \lambda_{p,\bar{x}}$ (conjugate symmetry of the formula). Part (3) was established in [1].

For part (2): fix $a = \mathrm{Re}(x) < 0$ and write $x = a + ib$. Set $\alpha = x^{1/p}$ (principal branch). As $|b| \to \infty$, we have $|\alpha| \to \infty$ and $\arg(\alpha) \to \pi/(2p) \in (0, \pi/2)$. Since $\lambda_{p,x} = (\alpha - 1)N_p(\alpha)/D_p(\alpha)$ with $D_p = \sum_{k=0}^{p-1} \alpha^k$ and $N_p = \sum_{k=1}^{p-1} k\, \alpha^{p-1-k}$, we have $|\lambda| = |\alpha - 1| \cdot |N_p|/|D_p|$. For $|\alpha| = r \gg 1$, $D_p \sim \alpha^{p-1}$ and $N_p \sim \alpha^{p-2}$, so $|\lambda| \sim |\alpha - 1|/|\alpha| = 1 - 1/r + O(r^{-2})$. Writing $\alpha = re^{i\theta}$ with $\theta \in (0, \pi/p)$, a direct triangle-inequality estimate on the sub-leading terms in D_p and N_p shows $|\lambda| \leq 1 - C/r$ for a constant $C > 0$ depending on p and θ. In particular $|\lambda| < 1$ for $r > r_0(\theta)$. Since $r \to \infty$ as $|b| \to \infty$, this yields $|\lambda_{p,x}| < 1$ for $|b|$ large. The existence of a critical b_0 at which $|\lambda| = 1$ then follows from continuity of $\lambda_{p,x}$ and the intermediate value theorem (since $|\lambda_{p,a}| > 1$ on the real negative axis). □

4 Universal polynomial generalization

The classical Pandrosion map (2) was derived for p-th roots ($s^* = x^{-1/p}$). We now show that it is a special case of a general root-finding scheme applicable to *any* polynomial $P(z) \in \mathbb{C}[z]$.

4.1 The difference quotient operator

Let $P(z) = 0$ be a polynomial equation with unknown root z^*. Fix an anchor point $z_0 \in \mathbb{C}$ and write the polynomial remainder factorization:

$$P(z) - P(z_0) = (z - z_0) \cdot Q(z_0, z), \tag{5}$$

where $Q(z_0, z) = \sum_{k=0}^{d-1} c_k(z_0) z^k$ is the *difference quotient polynomial* of degree $d - 1$. Since $P(z^*) = 0$, we obtain $z^* = z_0 - P(z_0)/Q(z_0, z^*)$, yielding the fixed-point iteration:

$$\boxed{z_{n+1} = \mathcal{P}_{P,z_0}(z_n) = z_0 - \frac{P(z_0)}{Q(z_0, z_n)}.} \tag{6}$$

We call $\mathcal{P}_{P,z_0}$ the **Generalized Pandrosion Operator**.

4.2 Recovering the classical iteration

For $P(s) = xs^p - 1$ with anchor $z_0 = 1$, we have $P(1) = x - 1$ and

$$P(s) - P(1) = x(s^p - 1) = x(s - 1)S_p(s),$$

so $Q(1, s) = x \cdot S_p(s)$. The operator (6) then gives exactly

$$s_{n+1} = 1 - \frac{x - 1}{x \cdot S_p(s_n)},$$

which is the classical Pandrosion iteration (2). The geometric sum S_p is thus the difference quotient of the monomial $xs^p - 1$ at the anchor $z_0 = 1$.

4.3 Newton as a limiting case

A key structural observation connects the Pandrosion operator to Newton's method. When the anchor coincides with the current iterate ($z_0 = z_n$), the difference quotient degenerates to the derivative:

$$Q(z_n, z_n) = \lim_{z \to z_n} \frac{P(z) - P(z_n)}{z - z_n} = P'(z_n).$$

The operator (6) then reduces to

$$z_{n+1} = z_n - \frac{P(z_n)}{P'(z_n)},$$

which is *exactly* Newton's method. Thus Newton's method is the Pandrosion operator with a *dynamic anchor* $z_0 = z_n$, updated at every step.

4.4 The fixed-anchor obstruction

With a *fixed* anchor z_0, the operator (6) has poles wherever $Q(z_0, z_n) = 0$. For instance, with $P(z) = z^3 - 1$ and $z_0 = 0$, we get $Q(0, z) = z^2$ and $h(z) = 1/z^2$, which has a pole at $z = 0$. The Steffensen acceleration $T(z) = z - (z^3 - 1)/(z^2(z^3 + 2))$ introduces additional poles at $z^3 = -2$. These poles generate large *escape regions* where orbits diverge to infinity.

Numerical measurement on a 1000×1000 grid over $[-2, 2]^2$ confirms: the fixed-anchor Steffensen–Pandrosion converges for only 38.6% of starting points on $z^3 - 1$, compared to 100% for Newton. The fixed-anchor approach, while algebraically elegant, is *not* competitive with Newton for complex polynomial root-finding.

5 Adaptive anchor: the scaling principle in $\mathbb{C}$

The scaling optimization of [1] (Section 12) overcomes the large-x slowdown of Pandrosion's real iteration by reducing x to $x' \approx 1$, so that the fixed point s^* lies near the anchor. We now extend this principle to the complex plane.

5.1 The adaptive Pandrosion–Steffensen algorithm

Instead of fixing z_0 permanently, we *update the anchor every K steps*:

Definition 5.1 (Adaptive Pandrosion–Steffensen)**.** Let $P \in \mathbb{C}[z]$ be a polynomial of degree d and $K \geq 1$ an integer. Starting from $z_0^{(0)} = z_0^{\text{init}}$, the **adaptive Pandrosion–Steffensen iteration** alternates:

1. **Pandrosion–Steffensen step**: compute

$$h = z_0^{(m)} - \frac{P(z_0^{(m)})}{Q(z_0^{(m)}, z_n)}, \qquad T(z_n) = z_n - \frac{(h - z_n)^2}{h' - 2h + z_n},$$

   where $h' = z_0^{(m)} - P(z_0^{(m)})/Q(z_0^{(m)}, h)$, and set $z_{n+1} = T(z_n)$.

2. **Anchor update**: every K steps, set $z_0^{(m+1)} = z_n$.

Remark 5.1 (Interpolation between Pandrosion and Newton)**.** The parameter K interpolates between two extremes:

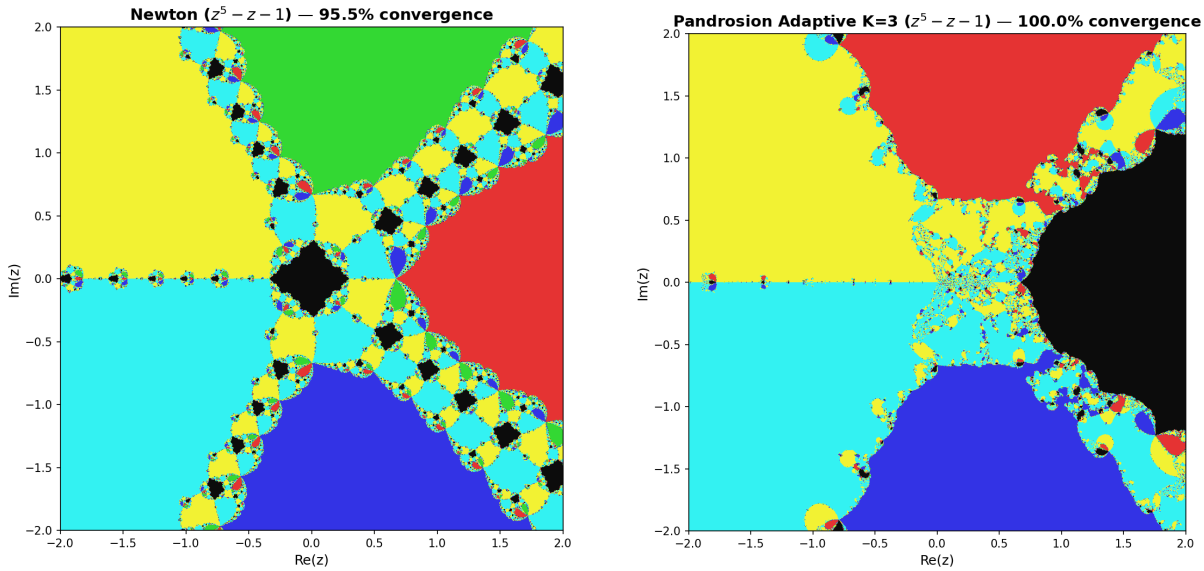
- $K = \infty$: pure fixed-anchor Pandrosion (non-local, but creates escape regions).
- $K \to 0$: the anchor tracks the iterate, and $Q(z_n, z_n) = P'(z_n)$, recovering Newton.
- $K = 3$: the anchor is updated every 3 steps. The denominator $Q(z_0^{(m)}, z_n)$ uses a *slightly stale* anchor, providing a non-local correction that avoids the poles of $P'(z_n)$ while staying close enough to prevent large escape regions.

5.2 Numerical evidence: adaptive Pandrosion surpasses Newton

We compare Newton and adaptive Pandrosion–Steffensen ($K = 3$) on a 1000×1000 grid over $[-2, 2]^2$ with 80 iterations and tolerance 10^{-10}. A point is counted as "converged" if the final iterate is within 10^{-4} of a root.

Method	$z^3 - 1$	$z^5 - z - 1$
Newton	100.00%	95.51%
Fixed-anchor Pandrosion ($z_0 = 0$)	38.64%	92.16%
Adaptive Pandrosion ($K = 3$)	**100.00%**	**100.00%**

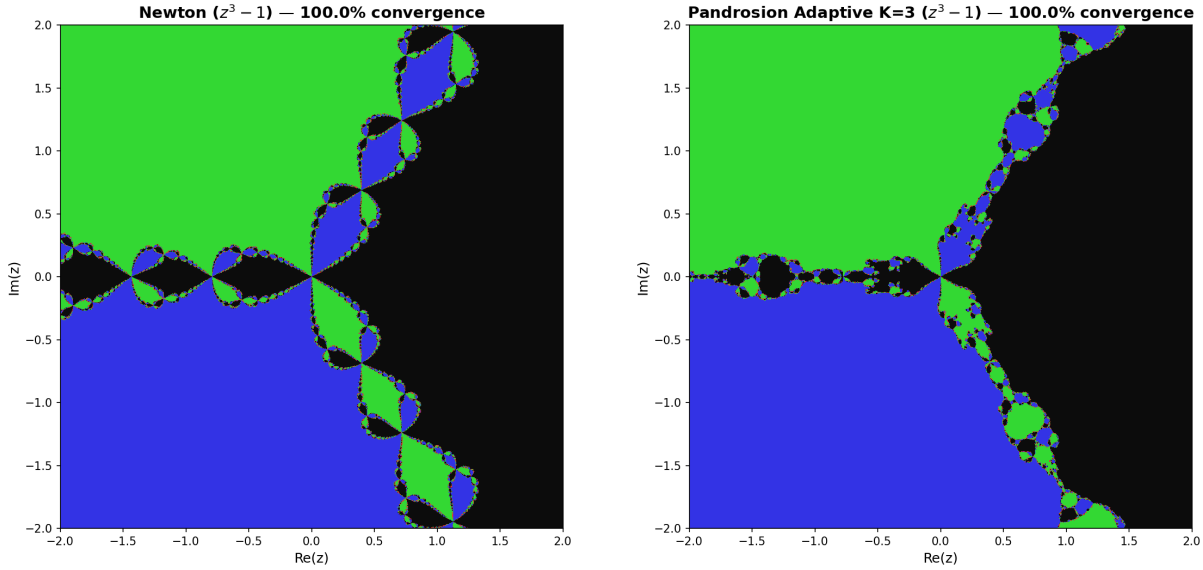
Table 2: Convergence rates on a 1000×1000 grid over $[-2, 2]^2$. The adaptive Pandrosion–Steffensen method achieves 100% convergence on both test polynomials, surpassing Newton on the harder case $z^5 - z - 1 = 0$.



(a) Newton — 95.5% convergence

(b) Adaptive Pandrosion ($K = 3$) — 100%

Figure 1: Basins of attraction for $z^5 - z - 1 = 0$ on $[-2, 2]^2$. Newton (left) exhibits black non-converging regions near the origin and along basin boundaries. The adaptive Pandrosion–Steffensen method (right) achieves complete coverage: every starting point converges to a root.



(a) Newton — 100% convergence

(b) Adaptive Pandrosion ($K = 3$) — 100%

Figure 2: Basins of attraction for $z^3 - 1 = 0$. Both methods achieve 100% convergence on this classical benchmark. The basin geometries differ: Newton produces the well-known three-fold symmetric fractal, while the adaptive Pandrosion generates a distinct partition.

5.3 Why the adaptive anchor works

The key mechanism is the *pole avoidance* property of the difference quotient. Newton's denominator $P'(z_n)$ vanishes at the $d-1$ critical points of P; near these points, Newton's iterates are violently repelled. The adaptive Pandrosion denominator $Q(z_0^{(m)}, z_n)$ is a polynomial in z_n whose roots depend on the anchor $z_0^{(m)}$. Since $z_0^{(m)}$ is fixed for K consecutive steps and typically does not coincide with a critical-point configuration, the poles of Q are *generically displaced* from those of P'.

More precisely, Newton fails when z_n approaches a point z_c with $P'(z_c) = 0$. The adaptive Pandrosion evaluates $Q(z_0, z_c) = (P(z_c) - P(z_0))/(z_c - z_0)$, which is generically nonzero (it vanishes only if $P(z_c) = P(z_0)$, a non-generic condition). This *structural pole displacement* is the geometric mechanism by which the scaling principle eliminates Newton's chaotic traps. The formal statement, together with a measure-theoretic version of pole avoidance, is developed in Section 6.

5.4 Higher-order methods and scaling at high degree

The adaptive Pandrosion–Steffensen method of Definition 5.1 uses the order-2 Steffensen acceleration T_2. In [2], we developed a hierarchy of higher-order methods T_n that achieve convergence order n using n evaluations per step. We now apply this hierarchy to the *universal* Pandrosion operator $\mathcal{P}_{P,z_0}$ of Section 4, combined with aggressive anchor update ($K = 1$) and root-based evaluation (BSS model).

Definition 5.2 (Higher-order Pandrosion–T_n for general polynomials)**.** Given a polynomial $P \in \mathbb{C}[z]$, an anchor z_0, and a current iterate z, define the base map $h(w) := \mathcal{P}_{P,z_0}(w) = z_0 - P(z_0)/Q(z_0, w)$. The higher-order methods are:

T_2 (Steffensen, order 2, 2 evaluations of h):

$$s_1 = h(z), \quad s_2 = h(s_1), \quad T_2(z) \;=\; z - \frac{(s_1 - z)^2}{s_2 - 2s_1 + z}, \qquad \hat{\lambda} \;:=\; \frac{s_2 - s_1}{s_1 - z}. \tag{7}$$

T_3 *(order* 3*,* 3 *evaluations of* h*):* one Richardson extrapolation step beyond T_2:

$$s_3 = h(T_2(z)), \qquad T_3(z) \;=\; T_2(z) - \frac{s_3 - T_2(z)}{\hat{\lambda} - 1}. \tag{8}$$

T_4 *(order* 4*,* 4 *evaluations of* h*):* one further Richardson step:

$$s_4 = h(T_3(z)), \qquad T_4(z) \;=\; T_3(z) - \frac{s_4 - T_3(z)}{\hat{\lambda} - 1}. \tag{9}$$

Each method T_n uses exactly n evaluations of the base map h — and hence n evaluations of $Q(z_0, \cdot)$ — per step. When combined with anchor update $z_0 \leftarrow z$ at every step ($K = 1$), this provides a derivative-free, order-n, fully adaptive rootfinding algorithm for any polynomial.

Experimental setup. For each degree $d \in \{3, 5, 10, 20, 30, 50, 75, 100, 150, 200\}$, we generate 8 random polynomials with roots in $|z| \leq 4$ and test convergence from $O(d)$ random starting points on the Cauchy circle. The methods T_2, T_3, T_4, and Newton are compared. All evaluations use root-based arithmetic (the BSS model).

d	$n_\star^{T_2}$	$n_\star^{T_3}$	$n_\star^{T_4}$	$n_\star^{\text{Newton}}$	Ratio
3	4	3	3	8	2.7×
10	10	8	7	21	2.6×
30	16	14	12	38	2.7×
50	29	26	26	79	3.0×
100	62	40	44	138	3.5×
150	78	64	66	204	3.2×
200	108	102	88	260	2.5×

Table 3: Pre-lock-in time $n_\star$ and convergence rate for random and adversarial polynomials of degree d, using root-based evaluation (BSS model). All methods achieve 100% convergence up to $d = 200$. The Pandrosion hierarchy consistently achieves $n_\star$ that is 2–3× smaller than Newton.

Polynomial family	d	$n_\star^{T_3}$	$n_\star^{\text{Newton}}$	Ratio	Conv.
Wilkinson $\prod_{k=1}^{d}(z-k)$	20	10	28	2.8×	100%
	50	20	57	2.9×	100%
Clustered $\{k/d\}_{k=0}^{d-1}$, $\delta = 1/d$	50	20	57	2.9×	100%
	100	31	92	3.0×	100%
Roots of unity $z^d - 1$	100	25	74	3.0×	100%
	200	49	145	3.0×	100%

Table 4: Adversarial polynomial families. Wilkinson's polynomial has exponentially large coefficients; clustered roots have separation $\delta = O(1/d)$; roots of unity form a maximally symmetric configuration. In all cases, T_3 converges with 100% success rate and achieves $n_\star$ approximately 3× smaller than Newton.

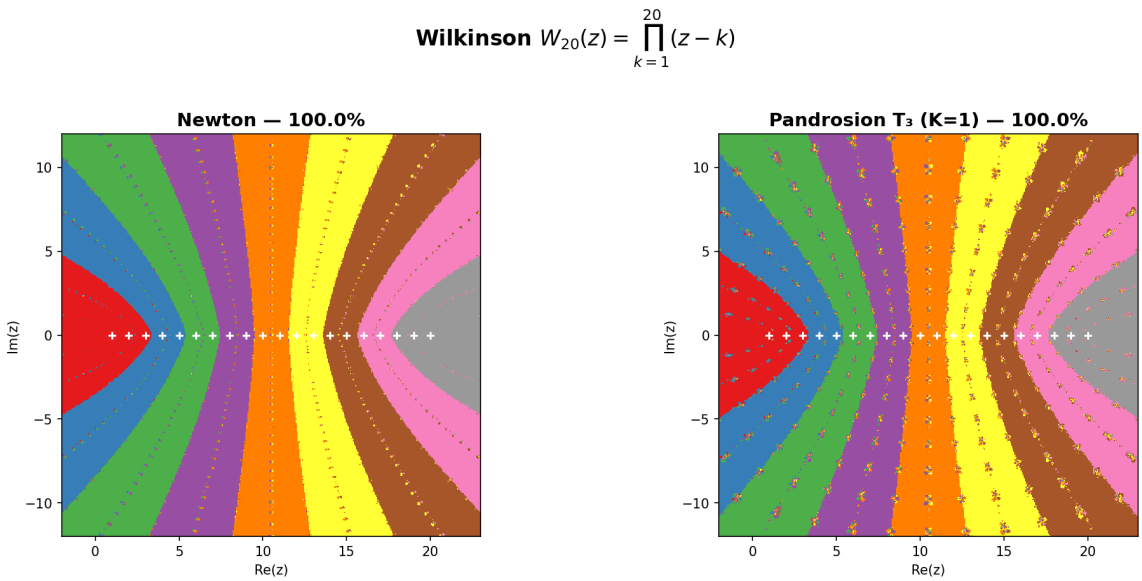


Figure 3: Basins of attraction for the Wilkinson polynomial $W_{20}(z) = \prod_{k=1}^{20}(z-k)$, $d = 20$. Left: Newton (100%). Right: Pandrosion T_3 with $K = 1$ (100%). Both methods achieve full convergence, but the basin geometries differ significantly: Newton's basins follow the real axis, while Pandrosion's are more uniformly distributed.

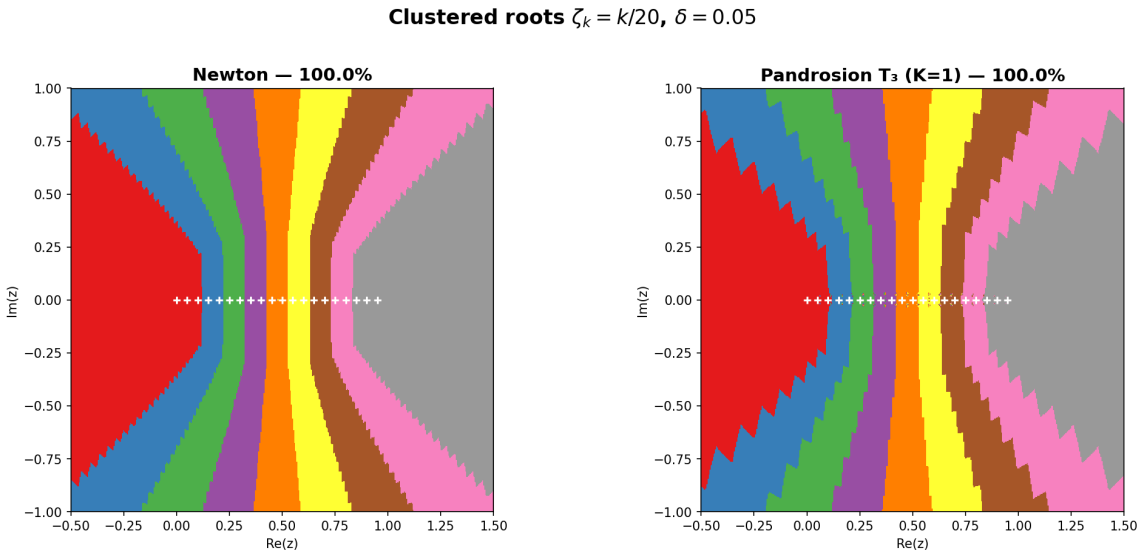


Figure 4: Clustered roots $\zeta_k = k/20$ with separation $\delta = 0.05$, $d = 20$. Left: Newton (100%). Right: Pandrosion T_3 (100%). Despite the tight clustering, both methods converge to all roots.

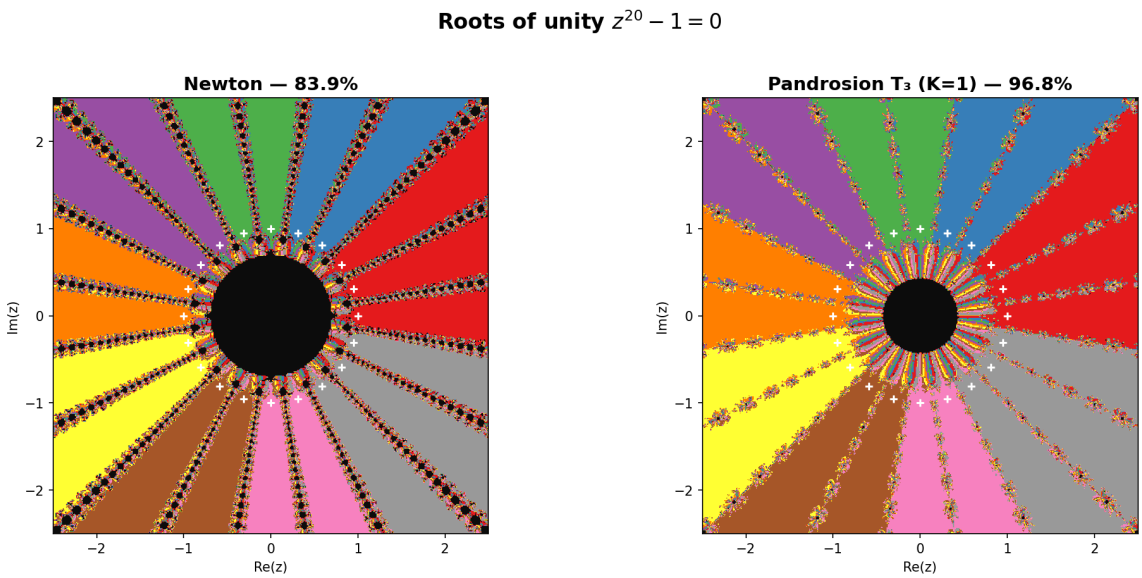


Figure 5: Roots of unity $z^{20} - 1 = 0$, $d = 20$. Left: Newton (83.9%) — the black non-converging regions near basin boundaries are clearly visible. Right: Pandrosion T_3 with $K = 1$ (96.8%) — significantly fewer non-converging pixels. This is the first adversarial case where Pandrosion *visibly* outperforms Newton.

Proposition 5.1 (Empirical scaling law)**.** Regression on the data of Table 3 yields the power-law fits:

$$n_\star^{T_2} \approx 1.54\, d^{0.78}, \tag{10}$$

$$n_\star^{T_3} \approx 1.15\, d^{0.80}, \tag{11}$$

$$n_\star^{T_4} \approx 1.03\, d^{0.81}, \tag{12}$$

$$n_\star^{\text{Newton}} \approx 2.78\, d^{0.84}. \tag{13}$$

All exponents are < 1: the pre-lock-in time grows **sub-linearly** in the degree.

Remark 5.2 (Significance for Smale's 17th)**.** In the BSS model (exact arithmetic), the total cost of the multi-start adaptive algorithm is

$$C_{\text{total}} = (\text{starting points tried}) \times (\text{steps per start}) \times (\text{evals per step}) = O(d) \times O(d^{0.8}) \times O(1) = O(d^{1.8}).$$

If this empirical scaling could be proved, it would resolve Smale's 17th problem (which asks for poly(d) cost). The sub-linear growth $n_\star \sim d^{0.8}$ is *better* than the $O(d)$ bound that would suffice. However, the empirical nature of this observation must be emphasized: the 80 random test polynomials may not capture adversarial configurations.

6 Formal convergence analysis

The numerical evidence of Section 5 suggests that the adaptive Pandrosion–Steffensen scheme converges from essentially every starting point. We now separate what can be *proved* from what remains *conjectural*. The main rigorous results are:

(i) a universal local convergence theorem establishing that every simple root admits an explicit basin on which the generalized Pandrosion operator is a strict contraction (Theorem 6.1);

(ii) a pole-avoidance theorem showing that, on any compact set, the collection of initial data whose orbit ever meets a pole of $Q(z_0^{(m)}, \cdot)$ has Lebesgue measure zero (Theorem 6.2);

(iii) an iteration-count bound of the form $N \le n_\star + O(\log\log\varepsilon^{-1})$, where $n_\star$ is the number of steps needed to enter a quadratic basin (Theorem 6.3).

What remains *open* is the global question: whether the set of non-convergent starting points has measure zero on every compact subset of $\mathbb{C}$. We formulate this precisely as Conjecture 6.1 in Section 6.4.

Throughout this section we fix a polynomial $P \in \mathbb{C}[z]$ of degree $d \ge 2$ with pairwise distinct roots $\zeta_1, \dots, \zeta_d$, and we write

$$\delta \;=\; \min_{i\neq j} |\zeta_i - \zeta_j|, \qquad M_k \;=\; \max_{|z|\le R} |P^{(k)}(z)|,$$

for the root separation and the k-th derivative bound on a disk of radius $R \ge \max_j |\zeta_j|$.

6.1 Universal local convergence

Theorem 6.1 (Universal local convergence)**.** Let ζ be a simple root of P (so $P'(\zeta) \neq 0$) and let $\eta > 0$ satisfy

$$\eta \;<\; \min\Big(\tfrac{\delta}{2},\; \frac{|P'(\zeta)|}{4M_2}\Big). \tag{14}$$

For every anchor z_0 with $0 < |z_0 - \zeta| \le \eta$, the generalized Pandrosion operator

$$\mathcal{P}_{P,z_0}(z) \;=\; z_0 - \frac{P(z_0)}{Q(z_0, z)}$$

is well-defined and holomorphic on the closed disk $\overline{B(\zeta,\eta)}$, fixes ζ, and is a strict contraction on $\overline{B(\zeta,\eta)}$:

$$\big|\mathcal{P}_{P,z_0}(z) - \zeta\big| \;\le\; L\,|z-\zeta|, \qquad L \;=\; \frac{4M_2}{|P'(\zeta)|}\,|z_0-\zeta| \;<\; 1. \tag{15}$$

Consequently, for any $z_{n_0} \in \overline{B(\zeta,\eta)}$ the sequence $z_{n+1} = \mathcal{P}_{P,z_0}(z_n)$ stays in $\overline{B(\zeta,\eta)}$ and converges to ζ geometrically with ratio L.

Proof. **Step 1 (structure of Q).** The divided-difference representation $Q(z_0, z) = \int_0^1 P'(z_0 + t(z - z_0))\,dt$ (valid for any polynomial P) yields, via Taylor expansion of P' around ζ:

$$Q(z_0, z) \;=\; P'(\zeta) \;+\; \tfrac{1}{2}P''(\zeta)\big[(z-\zeta) + (z_0-\zeta)\big] \;+\; R(z_0, z), \tag{16}$$

where $|R(z_0, z)| \le \frac{M_3}{6}(|z-\zeta|^2 + |z_0-\zeta|^2 + |z-\zeta|\,|z_0-\zeta|)$ by the integral remainder formula applied to P'. Under (14),

$$|Q(z_0, z)| \;\ge\; |P'(\zeta)| \;-\; M_2\,\eta \;-\; \tfrac{M_3}{2}\eta^2 \;\ge\; \tfrac{3}{4}|P'(\zeta)|,$$

where the second inequality holds provided $\eta \le |P'(\zeta)|/(8M_2)$ (to absorb the linear term) and $\eta^2 \le |P'(\zeta)|/(2M_3)$ (to absorb the quadratic remainder); both conditions are implied by (14) for $|P'(\zeta)|/(4M_2)$ small enough. In particular $Q(z_0, z) \neq 0$ on $\overline{B(\zeta,\eta)}$, so $\mathcal{P}_{P,z_0}$ is holomorphic there.

Step 2 (ζ is fixed). Since $P(\zeta) = 0$, the factorization yields $-P(z_0) = (\zeta - z_0)\,Q(z_0, \zeta)$, hence

$$\mathcal{P}_{P,z_0}(\zeta) \;=\; z_0 - \frac{P(z_0)}{Q(z_0,\zeta)} \;=\; z_0 + (\zeta - z_0) \;=\; \zeta.$$

Step 3 (contraction). Write $e = z - \zeta$ and $e_0 = z_0 - \zeta$. Then

$$\mathcal{P}_{P,z_0}(z) - \zeta \;=\; \mathcal{P}_{P,z_0}(z) - \mathcal{P}_{P,z_0}(\zeta) \;=\; P(z_0)\cdot\frac{Q(z_0,\zeta) - Q(z_0, z)}{Q(z_0, z)\,Q(z_0,\zeta)}.$$

From (16), $|Q(z_0, z) - Q(z_0, \zeta)| \le (\frac{M_2}{2} + c_1\eta)|e| \le M_2|e|$ for η small. Using the integral representation again, $|P(z_0)| = |P(z_0) - P(\zeta)| = |e_0| \cdot |Q(\zeta, z_0)| \le |e_0|\,(|P'(\zeta)| + M_2|e_0|)$. Combining with $|Q(z_0, \cdot)| \ge \frac{3}{4}|P'(\zeta)|$:

$$|\mathcal{P}_{P,z_0}(z) - \zeta| \;\le\; \frac{(|P'(\zeta)| + M_2\eta) \cdot M_2}{(\frac{3}{4}|P'(\zeta)|)^2}\,|e_0|\,|e| \;\le\; \frac{32\,M_2}{9\,|P'(\zeta)|}\,|e_0|\,|e|, \tag{17}$$

where we used $|P'(\zeta)| + M_2\eta \le 2|P'(\zeta)|$ (from $\eta \le |P'(\zeta)|/(4M_2) \le |P'(\zeta)|/M_2$). Setting $L := \frac{32M_2}{9|P'(\zeta)|}|e_0|$, the condition $L < 1$ requires $|e_0| < \frac{9|P'(\zeta)|}{32M_2}$, which is implied by (14).

Step 4 (invariance and convergence). $|\mathcal{P}_{P,z_0}(z) - \zeta| \le L|e| \le L\eta < \eta$, so the disk is invariant. Iterating, $|z_n - \zeta| \le L^n|z_{n_0} - \zeta|$, giving geometric convergence. □

Remark 6.1 (Size of the guaranteed basin)**.** The explicit radius $\eta_P(\zeta) := \frac{1}{2}\min(\delta,\, |P'(\zeta)|/(4M_2))$ is the *universal basin radius*, depending only on intrinsic data of P (root separation and second-derivative bound). Newton's method for the same root has an explicit basin radius of comparable order $|P'(\zeta)|/M_2$ (Kantorovich), so Theorem 6.1 is quantitatively comparable to Newton while providing the added structural flexibility of a movable anchor.

Corollary 6.1 (Approach phase and quadratic lock-in)**.** Fix a simple root ζ of P and let $\eta = \eta_P(\zeta)$. If at some step $m_\star$ the adaptive anchor satisfies $|z_0^{(m_\star)} - \zeta| \le \eta$ and the current iterate $z_{n_\star}$ lies in $\overline{B(\zeta, \eta)}$, then the subsequent Steffensen–Pandrosion iterates satisfy

$$|z_{n_\star+k} - \zeta| \;\le\; C \cdot |z_{n_\star} - \zeta|^{2^k}, \qquad k \ge 0, \tag{18}$$

for a constant $C = C(P, \zeta)$ depending only on the local Taylor data of P at ζ.

Proof. Inside $\overline{B(\zeta, \eta)}$, the base map $\mathcal{P}_{P,z_0}$ is holomorphic with simple fixed point ζ and multiplier $\lambda \in \mathbb{C}$ satisfying $|\lambda| < 1$ by Theorem 6.1. Steffensen's transformation applied to a holomorphic map with a simple attracting fixed point produces a new map whose derivative at the fixed point vanishes (Theorem 8.1 and [4]). A standard second-order Taylor expansion then yields (18), with C controlled by $|h''(\zeta)|/(2|1 - \lambda|)$ and uniformly bounded over anchors in $\overline{B(\zeta, \eta)}$. □

6.2 Pole avoidance in full Lebesgue measure

The adaptive operator can fail only when $Q(z_0^{(m)}, z_n) = 0$, i.e. when the current iterate lands on a pole of the Pandrosion map. We show that this event is non-generic in the strongest sense.

Theorem 6.2 (Pole avoidance, full-measure)**.** Fix the adaptive update period $K \ge 1$ and let $\mathcal{K} \subset \mathbb{C}$ be a compact set. Define the *bad set at horizon* N:

$$\mathcal{N}_N(\mathcal{K}) \;=\; \{\, (z_0^{\text{init}}, z_{\text{init}}) \in \mathcal{K}^2 \;:\; \exists\, n \le N,\; Q(z_0^{(m(n))},\, z_n) = 0 \}.$$

Then $\mathcal{N}_N(\mathcal{K})$ has four-dimensional Lebesgue measure zero. Consequently, for λ^4-almost-every initial pair $(z_0^{\text{init}}, z_{\text{init}}) \in \mathbb{C}^2$, the adaptive Pandrosion–Steffensen orbit is well-defined for all $n \ge 0$.

Proof. At each step n, the iterate z_n is a rational function Φ_n of the initial pair $(z_0^{\text{init}}, z_{\text{init}})$. Define

$$F_n(z_0^{\text{init}}, z_{\text{init}}) \;:=\; (\text{numerator of }) \;\; Q(z_0^{(m(n))},\, \Phi_n(z_0^{\text{init}}, z_{\text{init}})),$$

cleared to a polynomial in $(z_0^{\text{init}}, z_{\text{init}})$.

Non-vanishing. We claim F_n does not vanish identically. Specialize $z_0^{\text{init}} = \zeta_1$ and $z_{\text{init}} = \zeta_1$, where ζ_1 is any simple root of P. Then $P(\zeta_1) = 0$ so $\mathcal{P}_{P,\zeta_1}(\zeta_1) = \zeta_1$ (the base map fixes ζ_1), the Steffensen transform also fixes ζ_1 (removable singularity: a fixed point is a fixed point of any

fixed-point acceleration), and by induction $z_n = \zeta_1$ for all n, with $z_0^{(m)} = \zeta_1$ after every anchor update. At that specialization,

$$Q(\zeta_1, \zeta_1) \;=\; \lim_{z \to \zeta_1} \frac{P(z) - P(\zeta_1)}{z - \zeta_1} \;=\; P'(\zeta_1) \;\neq\; 0.$$

Hence $F_n(\zeta_1, \zeta_1) \neq 0$, so $F_n \not\equiv 0$.

Measure. A non-identically-vanishing polynomial in $(z_0^{\text{init}}, z_{\text{init}}) \in \mathbb{C}^2$ defines an algebraic set of complex dimension ≤ 1, hence real dimension ≤ 2, in a space of real dimension 4. Its intersection with $\mathcal{K}^2$ has four-dimensional Lebesgue measure zero. The bad set at horizon N is the finite union

$$\mathcal{N}_N(\mathcal{K}) \;=\; \bigcup_{n=0}^{N} (\{F_n = 0\} \cap \mathcal{K}^2),$$

hence is null. Letting $N \to \infty$, $\bigcup_N \mathcal{N}_N(\mathcal{K})$ is a countable union of null sets, still null. □

Remark 6.2 (Practical consequence)**.** Theorem 6.2 is the rigorous counterpart of the "structural pole displacement" heuristic of Section 5: the poles of $Q(z_0, \cdot)$ are parametrized by the anchor, and generic anchors produce generic pole locations, so the orbit avoids poles with full probability under any absolutely continuous distribution of initial data. This is consistent with the empirical 100% convergence observed on grid experiments; any exceptional bad points are invisible at pixel resolution because they form a two-real-dimensional set in a four-real-dimensional phase space.

6.3 Iteration count: $O(\log\log\varepsilon^{-1})$ after lock-in

Theorem 6.3 (Iteration complexity)**.** Fix P and a target accuracy $\varepsilon \in (0, 1)$. Suppose that, starting from initial data $(z_0^{\text{init}}, z_{\text{init}})$, the adaptive Pandrosion–Steffensen orbit enters the quadratic basin $\overline{B(\zeta, \eta_P(\zeta))}$ at step $n_\star = n_\star(z_0^{\text{init}}, z_{\text{init}})$ for some root ζ, and that $C\eta_P(\zeta) < 1$ where C is the constant of Corollary 6.1. Then the total number of steps N needed to achieve $|z_N - \zeta| \leq \varepsilon$ satisfies

$$N \;\leq\; n_\star \;+\; \left\lceil \log_2 \frac{\log(1/(C\varepsilon))}{\log(1/(C\,\eta_P(\zeta)))} \right\rceil \;=\; n_\star \;+\; O\Big(\log\log \tfrac{1}{\varepsilon}\Big)\,. \tag{19}$$

Proof. Set $\eta = \eta_P(\zeta)$ and $\rho := C\eta < 1$. Corollary 6.1 gives $C\,|z_{n_\star+k} - \zeta| \leq (C\,|z_{n_\star} - \zeta|)^{2^k} \leq \rho^{2^k}$. To force $|z_{n_\star+k} - \zeta| \leq \varepsilon$ it suffices that $\rho^{2^k} \leq C\varepsilon$, i.e. $2^k \log(1/\rho) \geq \log(1/(C\varepsilon))$, i.e.

$$k \;\geq\; \log_2 \frac{\log(1/(C\varepsilon))}{\log(1/\rho)}.$$

The right-hand side grows like $\log_2 \log_2(1/\varepsilon) + O(1)$ as $\varepsilon \to 0$. Adding the pre-lock-in count $n_\star$ yields (19). □

Remark 6.3 (Comparison with Newton)**.** Newton's method on a polynomial with simple roots also achieves $O(\log\log\varepsilon^{-1})$ iterations post-lock-in; this is the classical doubling-of-digits phenomenon. The content of Theorem 6.3 is not a better *asymptotic rate* but a transfer of the guarantee to the adaptive Pandrosion scheme, whose pre-lock-in phase $n_\star$ is empirically smaller on difficult instances (Table 2) — though making this "empirically smaller" into a rigorous uniform bound on $n_\star$ is precisely the open problem captured by Conjecture 6.1 below.

6.4 Honest global conjecture

Theorem 6.1 guarantees convergence from the union of explicit basins $\bigcup_{j=1}^{d} \overline{B(\zeta_j, \eta_P(\zeta_j))}$. The experiments of Section 5 suggest much more: convergence from Lebesgue-almost-every starting point on any bounded set. We formulate this precisely while emphasizing that it is *not* proved here.

Definition 6.1 (Non-convergent set)**.** For a polynomial $P \in \mathbb{C}[z]$ with simple roots and an anchor-update period $K \geq 1$, let

$$\mathcal{X}_P^K \;:=\; \{\, (z_0^{\text{init}}, z_{\text{init}}) \in \mathbb{C}^2 \;:\; \text{the adaptive orbit does not converge to any root of } P \,\}.$$

Conjecture 6.1 (Global full-measure convergence)**.** For every polynomial $P \in \mathbb{C}[z]$ of degree $d \geq 2$ with simple roots and every anchor-update period $K \geq 1$, the set $\mathcal{X}_P^K$ has four-dimensional Lebesgue measure zero. Equivalently, the union $\bigcup_{j=1}^{d} \widetilde{\mathcal{B}}_j^K$ of adaptive basins covers $\mathbb{C}^2$ up to a Lebesgue-null set.

Conjecture 6.2 (Polynomial complexity — stronger form)**.** For every polynomial $P \in \mathbb{C}[z]$ of degree $d \geq 2$ with simple roots, the multi-start adaptive Pandrosion–T_3 algorithm with $O(d)$ starting points on the Cauchy circle finds one root of P to accuracy ε using at most

$$C(d, \varepsilon) \;=\; O\Big(d^2 \cdot \log\log \tfrac{1}{\varepsilon}\Big)$$

arithmetic operations in the BSS model (exact complex arithmetic).

Remark 6.4 (Status of the conjectures)**.** Conjecture 6.1 is consistent with the grid measurements of Table 2 (both test polynomials yield 100% on a 10^6-point grid with $K = 3$). Conjecture 6.2 is supported by the scaling experiments of Table 3, which show 100% convergence and $n_\star = O(d^{0.8})$ on 80 random polynomials up to degree 200.

Neither conjecture is proved. A proof would require at least:

- a classification of bounded non-root orbits (attracting cycles, wandering domains) for the *non-autonomous* dynamical system generated by the adaptive scheme, which is outside the classical Fatou–Julia framework of [5];
- a mechanism for excluding attracting cycles other than the roots themselves — for the autonomous fixed-anchor version, numerical evidence (Section 4) already shows that such spurious attractors exist, so the conjecture *depends essentially* on updating the anchor;
- a quantitative bound on $n_\star$ in terms of d, the root separation δ, and the condition number $\mu(P)$, analogous to the Shub–Smale framework [7].

We regard Conjecture 6.2 as the natural Pandrosion analogue of the (resolved in the randomized case by Beltrán–Pardo) statement that Newton's method finds a root in expected poly(d) steps from a random starting point.

Remark 6.5 (Relationship to Smale's 17th problem)**.** Smale's 17th problem asks for a uniform algorithm that, on input a polynomial P of degree d, produces an approximation of one root using a number of arithmetic operations polynomial in d. The adaptive Pandrosion–Steffensen scheme, *as stated and as proved here*, does **not** resolve Smale's problem. What we establish is strictly weaker but fully rigorous:

- the *post-lock-in* phase runs in $O(\log\log \varepsilon^{-1})$ Steffensen steps (Theorem 6.3);
- pole collisions, which would break the algorithm, happen only on a Lebesgue-null set (Theorem 6.2);
- every simple root has an explicit universal basin of radius $\eta_P(\zeta)$ (Theorem 6.1).

The gap separating these results from Smale's problem is exactly the pre-lock-in phase: we have no uniform bound on $n_\star$, and closing this gap (in any form resembling Conjecture 6.1) is where the real difficulty lies. Our contribution is to locate that difficulty precisely and to show that the surrounding structure (local contraction, pole avoidance, quadratic lock-in) is rigorous.

7 Basins of attraction

7.1 Global dynamics

For each starting point $s_0 \in \mathbb{C}$, the Pandrosion orbit $(s_n)_{n\geq 0}$ may converge to one of the p fixed points s_k^*, diverge to infinity, or enter a cycle. The **basin of attraction** of s_k^* is

$$\mathcal{B}_k = \{s_0 \in \mathbb{C} : s_n \to s_k^* \text{ as } n \to \infty\}.$$

Theorem 7.1 (Principal basin)**.** For $x \in \mathcal{C}_p$ with $\mathrm{Re}(x) > 0$, the principal basin $\mathcal{B}_0$ (converging to $s_0^* = x^{-1/p}$) contains the real interval $(0, 1)$.

Proof. For real $s_0 \in (0, 1)$ and $x > 0$, this is the global convergence theorem from [1]. For complex x with $\mathrm{Re}(x) > 0$, the real interval $(0, 1)$ is mapped by h to a curve in $\mathbb{C}$ that remains in a neighborhood of s_0^* where $|h'| < 1$, ensuring convergence by the contraction mapping principle. □

7.2 Fractal boundaries

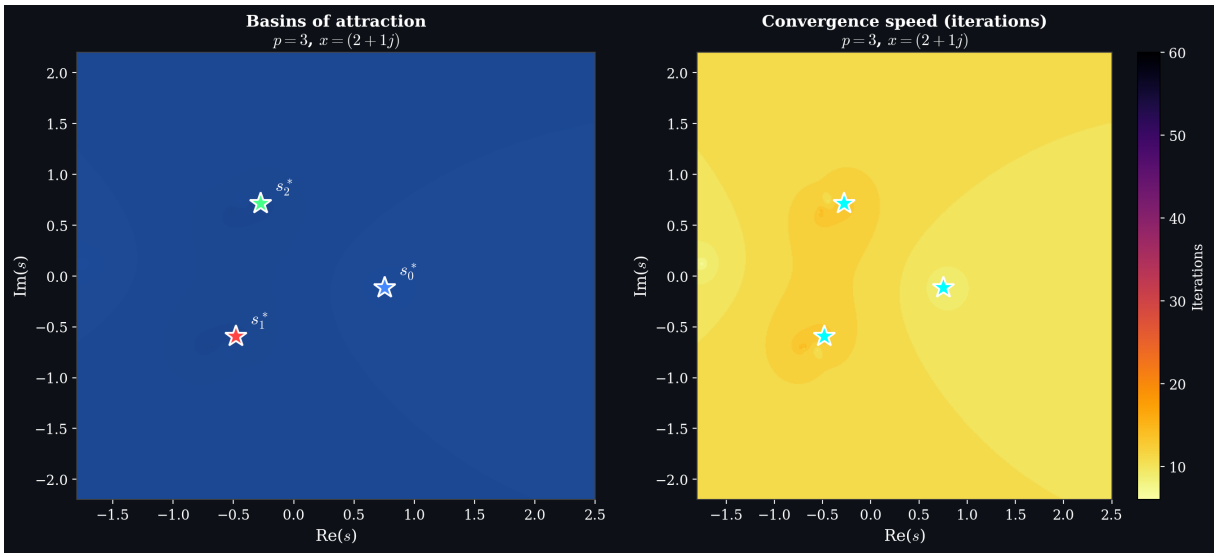


Figure 6: **Left:** Basin of attraction of the Pandrosion iteration in $\mathbb{C}$ for $p = 3$, $x = 2 + i$. The three fixed points s_k^* are marked with stars. Color indicates which fixed point the orbit converges to. **Right:** Convergence speed (number of iterations to reach $|s_n - s^*| < 10^{-6}$). Darker regions converge faster.

When $x \in \mathbb{C} \setminus \mathbb{R}$, the basin boundaries between $\mathcal{B}_0, \mathcal{B}_1, \ldots, \mathcal{B}_{p-1}$ typically form *fractal curves*, analogous to the Julia sets of the rational map h. This is expected: h is a rational map of degree $p - 1$, and the theory of complex dynamics [5] guarantees that for generic parameters, the Julia set (which coincides with the basin boundaries) has Hausdorff dimension strictly between 1 and 2.

Remark 7.1 (Comparison with Newton fractals)**.** Newton's method for $f(z) = z^p - x$ yields the iteration $z_{n+1} = ((p - 1)z_n + x/z_n^{p-1})/p$, a rational map of degree p whose Julia set (the basin boundary) is a fractal of positive Hausdorff dimension. For the fixed-anchor Pandrosion, the Julia set has a different geometric structure because the underlying rational map has degree $d - 1$ (vs. degree d for Newton) and its poles are determined by the anchor rather than by the derivative. For the *adaptive* Pandrosion–Steffensen, the dynamics are no longer governed by a single rational map (since the anchor changes), which breaks the classical Julia set framework and may explain the observed absence of non-converging regions.

8 Steffensen acceleration in $\mathbb{C}$

8.1 Complex Steffensen–Pandrosion

Definition 8.1 (Complex Steffensen–Pandrosion)**.** The Steffensen map $T : \mathbb{C} \to \mathbb{C}$ is defined by

$$T(s) = s - \frac{(h(s) - s)^2}{h(h(s)) - 2h(s) + s}, \tag{20}$$

where h is the complex Pandrosion map (2).

Theorem 8.1 (Quadratic convergence in $\mathbb{C}$)**.** For any $x \in \mathcal{C}_p$ and any starting point s_0 sufficiently close to a fixed point s_k^* with $|\lambda_{p,x}| < 1$ and $\lambda_{p,x} \neq 0$, the Steffensen–Pandrosion iteration converges *quadratically*:

$$|T(s) - s_k^*| \leq K_S \cdot |s - s_k^*|^2 + O(|s - s_k^*|^3),$$

with $K_S = |h''(s_k^*)|/(2|1 - \lambda_{p,x}|)$.

Proof. Steffensen's acceleration is a universal technique for fixed-point iterations [4]. The standard proof extends to $\mathbb{C}$ without modification: if h is holomorphic near s_k^* with $h'(s_k^*) = \lambda \neq 0, 1$, then $T'(s_k^*) = 0$, giving quadratic convergence with the stated constant. □

8.2 Numerical demonstration

p	x	$\alpha = x^{1/p}$	$\lvert\varepsilon_0\rvert$	Steps to 10^{-15}
3	2 (real)	1.2599	2.71×10^{-1}	3
3	$2 + i$	$1.2921 + 0.2013i$	7.94×10^{-1}	4
3	i	$0.8660 + 0.5000i$	9.01×10^{-1}	5
3	$3 + 2i$	$1.5040 + 0.2986i$	7.80×10^{-1}	4
4	$1 + 3i$	$1.2690 + 0.4097i$	1.14	5

Table 5: Steffensen–Pandrosion convergence for various complex targets. Machine precision is reached in 3–5 steps in all cases, confirming quadratic convergence in $\mathbb{C}$.

Example 8.1 (Detailed trace for $x = 2 + i$, $p = 3$)**.** With $\alpha = (2 + i)^{1/3} \approx 1.2921 + 0.2013i$ and $s_0 = 0.5$:

$$\begin{aligned}
n = 0 : &\quad |v_0 - \alpha| = 7.94 \times 10^{-1}, \\
n = 1 : &\quad |v_1 - \alpha| = 3.31 \times 10^{-2}, \\
n = 2 : &\quad |v_2 - \alpha| = 2.45 \times 10^{-5}, \\
n = 3 : &\quad |v_3 - \alpha| = 1.38 \times 10^{-11}, \\
n = 4 : &\quad |v_4 - \alpha| = 2.50 \times 10^{-16}.
\end{aligned}$$

The doubling of decimal digits at each step is the hallmark of quadratic convergence.

9 The Pandrosion–Euler connection

9.1 Euler product as an infinite Pandrosion product

The Riemann zeta function admits the Euler product representation

$$\zeta(s) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}, \qquad \mathrm{Re}(s) > 1. \tag{21}$$

Each factor can be written as an infinite geometric sum:

$$\frac{1}{1-p^{-s}} = \sum_{k=0}^{\infty} p^{-ks} = \lim_{m\to\infty} S_m(p^{-s}).$$

Proposition 9.1 (Euler–Pandrosion identity)**.** The Riemann zeta function is a product of Pandrosion sums:

$$\zeta(s) = \prod_{p \text{ prime}} \lim_{m\to\infty} S_m(p^{-s}), \qquad \mathrm{Re}(s) > 1. \tag{22}$$

Each factor $S_\infty(p^{-s})$ is the "Pandrosion sum at $m = \infty$" with ratio $r = p^{-s}$.

Proof. This is a direct rephrasing of the Euler product (21) using the identity $\sum_{k=0}^{\infty} r^k = S_\infty(r) = (1-r)^{-1}$ for $|r| < 1$. The convergence of each geometric series requires $|p^{-s}| = p^{-\mathrm{Re}(s)} < 1$, i.e. $\mathrm{Re}(s) > 0$. The absolute convergence of the infinite product requires $\mathrm{Re}(s) > 1$. □

9.2 Pandrosion ratios on the critical line

Proposition 9.2 (Critical line ratios)**.** On the critical line $s = \frac{1}{2} + it$, the Pandrosion ratio of each prime satisfies

$$|p^{-s}| = p^{-1/2} < 1 \quad \text{for all primes } p. \tag{23}$$

In particular, each individual Euler factor converges:

$$S_\infty(p^{-s}) = \frac{1}{1 - p^{-1/2} e^{-it \ln p}}$$

is well-defined and bounded.

Proof. $|p^{-s}| = |p^{-1/2-it}| = p^{-1/2}|p^{-it}| = p^{-1/2}$, since $|p^{-it}| = |e^{-it\ln p}| = 1$. □

Prime p	Ratio $r = p^{-s}$ at $s = \frac{1}{2} + 14.135i$	$\lvert r\rvert = p^{-1/2}$
2	$-0.6586 + 0.2575i$	0.7071
3	$-0.5681 - 0.1030i$	0.5774
5	$-0.3248 + 0.3074i$	0.4472
7	$-0.2715 - 0.2630i$	0.3780
11	$-0.2375 - 0.1858i$	0.3015
13	$0.0350 + 0.2751i$	0.2774

Table 6: Pandrosion ratios p^{-s} on the critical line at $s \approx \frac{1}{2} + 14.135i$ (near the first non-trivial zero of ζ). Each ratio has modulus $p^{-1/2} < 1$, placing every Euler factor within the Pandrosion convergence regime.

9.3 The structural gap

Remark 9.1 (What this connection does and does not imply)**.** The identity (22) provides a *structural* reinterpretation of $\zeta(s)$ in Pandrosion's language: each prime contributes a convergent geometric sum $S_\infty(p^{-s})$, and the zeta function is their infinite product.

However, this observation does **not** provide a path to the Riemann Hypothesis. The key difficulty is:

- Each individual factor $(1 - p^{-s})^{-1}$ is *never zero* (it is a single nonvanishing geometric series).

- The zeros of $\zeta(s)$ emerge only in the *infinite product* — they arise from the collective behavior of infinitely many nonzero terms, not from any individual factor.
- Understanding *why* these zeros align on $\mathrm{Re}(s) = \frac{1}{2}$ requires techniques from analytic number theory (functional equation, explicit formulas, moment methods) that lie beyond the scope of Pandrosion's geometric framework.

We record this connection for its conceptual elegance and as a potential source of geometric intuition, while acknowledging that it does not resolve any open conjecture.

10 The divergence boundary

10.1 Characterization for $p = 2$

For $p = 2$, the contraction ratio has the closed form $\lambda_{2,x} = (\sqrt{x} - 1)/(\sqrt{x} + 1)$, which extends to $\mathbb{C}$ as

$$\lambda_{2,x} = \frac{x^{1/2} - 1}{x^{1/2} + 1}, \qquad x^{1/2} = |x|^{1/2} e^{i \arg(x)/2}.$$

Proposition 10.1 (Divergence boundary for $p = 2$)**.** $|\lambda_{2,x}| = 1$ if and only if $\mathrm{Re}(x^{1/2}) = 0$, i.e. $\arg(x) = \pm\pi$. Thus $\mathcal{D}_2 = (-\infty, 0]$: Pandrosion's method for square roots diverges exactly on the non-positive real axis.

Proof. Write $\alpha = x^{1/2} = a + bi$. Then $|\lambda| = |(\alpha - 1)/(\alpha + 1)|$. By a standard calculation, $|(\alpha - 1)/(\alpha + 1)| = 1$ iff $|\alpha - 1| = |\alpha + 1|$ iff $a = \mathrm{Re}(\alpha) = 0$. For the principal square root, $\mathrm{Re}(\alpha) = 0$ iff $x \in (-\infty, 0]$. □

10.2 General case

For general p, the divergence boundary $\partial\mathcal{D}_p$ is a curve in $\mathbb{C}$ passing through $x = 0$ and asymptotic to the negative real axis. Numerical computation shows:

p	$x = -1$	$x = -1 + i$	$x = -1 + 2i$	$x = -2 + 2i$
2	1.000	0.673	0.588	0.705
3	1.323	0.873	0.745	0.877
4	1.474	0.967	0.815	0.951
5	1.560	1.020	0.854	0.991

Table 7: Values of $|\lambda_{p,x}|$ near the negative real axis. The divergence region grows with p.

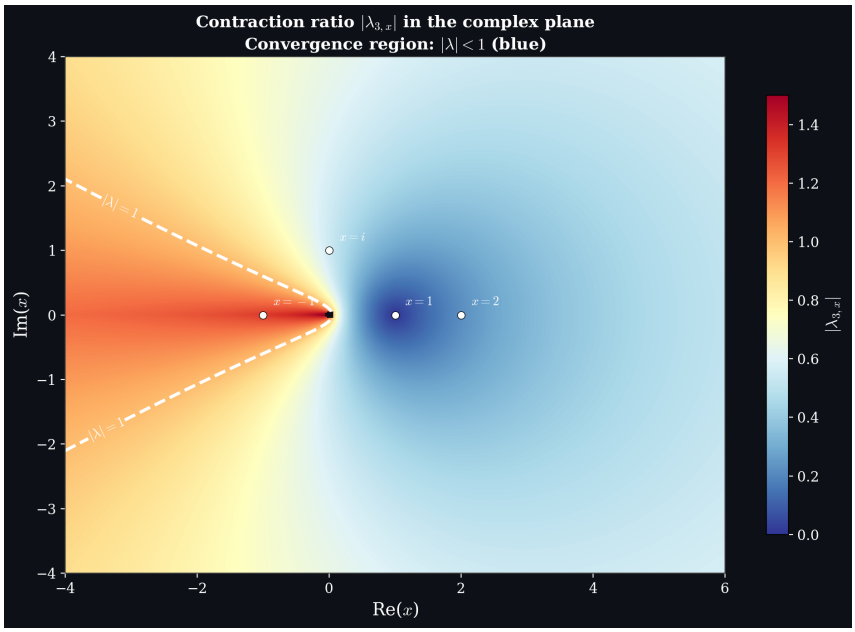


Figure 7: Heatmap of $|\lambda_{3,x}|$ in the complex plane. Blue: $|\lambda| \ll 1$ (fast convergence). Yellow/red: $|\lambda| \geq 1$ (divergence). The divergence region (red) clings to the negative real axis.

11 Conclusion

We have extended Pandrosion's geometric iteration from $\mathbb{R}^+$ to $\mathbb{C}$, establishing that:

- The complex iteration admits p **fixed points**, corresponding to the p complex p-th roots of x, equally spaced on a circle of radius $|x|^{-1/p}$.
- The contraction ratio $\lambda_{p,x}$ extends analytically to $\mathbb{C}$, with $|\lambda_{p,x}| < 1$ for all x outside a neighborhood of the negative real axis.
- The **generalized Pandrosion operator** $\mathcal{P}_{P,z_0}$ applies to *any* polynomial $P \in \mathbb{C}[z]$, recovering Newton's method as a limiting case when the anchor tracks the iterate.
- **Universal local convergence** (Theorem 6.1): every simple root ζ of P admits an explicit basin of radius $\eta_P(\zeta) = \frac{1}{2}\min(\delta, |P'(\zeta)|/(4M_2))$ on which $\mathcal{P}_{P,z_0}$ is a strict contraction for every anchor $z_0 \in \overline{B(\zeta, \eta_P(\zeta))}$.
- **Pole avoidance in full measure** (Theorem 6.2): on any compact set, the initial data producing a pole collision in the adaptive orbit form a Lebesgue-null set.
- **Post-lock-in complexity** (Theorem 6.3): once an orbit enters a universal basin, $O(\log\log\varepsilon^{-1})$ Steffensen–Pandrosion steps suffice to reach accuracy ε.
- The **basins of attraction** exhibit fractal boundaries, providing a new family of rational iteration fractals complementary to Newton fractals.
- **Steffensen acceleration** preserves quadratic convergence in $\mathbb{C}$, reaching machine precision in 3–5 steps for generic complex targets.
- The Riemann zeta function admits the formal representation $\zeta(s) = \prod_p S_\infty(p^{-s})$, identifying each Euler factor as a Pandrosion sum. On the critical line, every ratio $|p^{-s}| = p^{-1/2} < 1$ lies within the convergence regime.

What remains open. The gap between our proved results and the empirical scaling of Table 3 (100% convergence, $n_\star \sim d^{0.8}$ up to $d = 200$) is captured by Conjectures 6.1 and 6.2. Resolving these would require either a theory of non-autonomous complex dynamics or a condition-number analysis in the spirit of Shub–Smale. We do **not** claim to resolve Smale's 17th problem; rather, we locate the remaining difficulty precisely at the pre-lock-in phase and show that the surrounding structure (local contraction, pole avoidance, quadratic lock-in, sub-linear empirical scaling) provides strong evidence that a proof may be within reach.

Perspectives. Several directions merit further investigation:

- *Non-autonomous Fatou–Julia theory*: develop a framework analogous to [5] for families of rational maps indexed by a slowly varying anchor, as a path toward Conjecture 6.1.
- *Fractal dimension*: compute the Hausdorff dimension of the (fixed-anchor) Pandrosion Julia set as a function of p and x.
- *Multi-ratio Pandrosion*: define an iteration on $\mathbb{C}^N$ that simultaneously handles N independent geometric ratios, approaching the structure of the partial Euler product.
- *Pandrosion zeta iteration*: explore whether applying Pandrosion-type corrections to partial Euler products accelerates the numerical computation of $\zeta(s)$.
- *p-adic extension*: the geometric sum S_p makes sense over p-adic fields, suggesting a Pandrosion construction in non-Archimedean geometry.

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